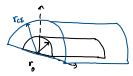
HW4

Wednesday, 19 April 2023 10:11 AM

Question 1

Hemicylinder electrode [WE's]



CE placed at a finite distance away from WE

(a) Assuming uniform composition:

$$\nabla^2\phi = 0$$

In cylindrical geometry: $\nabla^2\phi = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2\phi}{\partial\theta^2} + \frac{\partial^2\phi}{\partial z^2} = 0$

↳ Considering only the radial direction we get:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\phi}{\partial r}\right) = 0$$

BC's: $\phi = V$ @ $r = r_o$

$\phi = 0$ @ $r = r_c$

(b) Solve for ohmic resistance (R_Ω):

Obtaining ϕ profile: $\frac{\partial}{\partial r}\left(r\frac{\partial\phi}{\partial r}\right) = 0$

$$r\frac{\partial\phi}{\partial r} = c_1$$

$$\frac{\partial\phi}{\partial r} = \frac{c_1}{r}$$

$$\phi = c_1 \ln r + c_2$$

@ $r = r_o$, $\phi = V$ $\Rightarrow$ $V = c_1 \ln r_o + c_2$

@ $r = r_{CE}$, $\phi = 0$ $\Rightarrow$ $0 = c_1 \ln r_{CE} + c_2$

$$V = c_1 \ln\frac{r_o}{r_{CE}}$$

$$c_1 = \frac{V}{\ln\left(\frac{r_o}{r_{CE}}\right)} \Rightarrow$$

Solving for c_1: $0 = \frac{V}{\ln\left(\frac{r_o}{r_{CE}}\right)}\ln r_{CE} + c_2$

$$c_2 = -\frac{V}{\left(\ln\frac{r_o}{r_{CE}}\right)}\ln r_{CE}$$

$$\therefore \quad \phi = \frac{V}{\ln\left(\frac{r_o}{r_{CE}}\right)}\ln r - \frac{V}{\ln\left(\frac{r_o}{r_{CE}}\right)}\ln r_{CE}$$

$$\phi = \frac{V}{\ln\left(\frac{r_o}{r_{CE}}\right)}\cdot\ln\left(\frac{r}{r_{CE}}\right)$$

$R_\Omega = \frac{V}{i_\Omega}$ (Assuming no charge transfer resistances) $\Rightarrow$ $i_\Omega = -\kappa\left.\frac{\partial\phi}{\partial r}\right|_{r=r_o}$

$$\left.\frac{\partial\phi}{\partial r}\right|_{r=r_o} = \frac{c_1}{r_o} = \frac{V}{\ln\left(\frac{r_o}{r_{CE}}\right)}\cdot\frac{1}{r_o}$$

$$\therefore \quad R_\Omega = \frac{V}{-\kappa\frac{V}{r_o}\cdot\frac{1}{\ln\left(\frac{r_o}{r_{CE}}\right)}} = \boxed{\frac{r_o}{\kappa}\ln\left(\frac{r_{CE}}{r_o}\right) = R_\Omega}$$ ← In Ω

WE S.A.: $\pi r_o l$; l = length of cylinder

$$R'_\Omega = \frac{R_\Omega}{\pi r_o l} = \boxed{\frac{1}{\kappa\pi l}\ln\left(\frac{r_{CE}}{r_o}\right) = R'_\Omega}$$ ← In $\Omega\,cm^2$

As $r_{CE}\to\infty$, $R_\Omega = \frac{r_o}{\kappa}\ln\left(\frac{r_{CE}}{r_o}\right)$ ($\ln\left(\frac{r_{CE}}{r_o}\right)\to 0$)

$\lim_{r_{CE}\to\infty} R_\Omega = \infty$ ← If the CE is at an infinite distance away there is infinite ohmic resistance.

(c) For 1D: $R_\Omega = \frac{L}{\kappa}$ → This assumes that CE is at distance L

R_Ω for hemicylindrical assumes r_{CE} at finite distance away.

For hemispherical: $R_\Omega = \frac{r_o}{\kappa}$ → This assumes CE is at ∞

→ Comparing hemi-sphere to - cylinder

$\ln\left(\frac{r_{CE}}{r_o}\right)\to 0$ as $r_o\to\infty$

So cylindrical approaches spherical behaviour for large r_o

Question 2 [W 6.2]

Linear & Anodic Tafel kinetics

(a) $R_{ct} = \left.\frac{\partial \eta_s}{\partial i}\right|_{i=i_{avg}}$; $i = -i_o \exp\left(\frac{-\alpha_c F \eta_s}{RT}\right)$ → Cathodic Tafel ;

$$\frac{\partial i}{\partial \eta_s} = -i_o \left(\frac{-\alpha_c F}{RT}\right) \exp\left(\frac{-\alpha_c F \eta_s}{RT}\right)$$

$$= \frac{\alpha_c F \; i_o}{RT} \exp\left(-\frac{\alpha_c F \eta_s}{RT}\right)$$

@ $i = i_{avg}$

$$i_{avg} = -i_o \exp\left(-\frac{\alpha_c F \eta_s}{RT}\right)$$

$$\exp\left(\frac{-\alpha_c F}{RT}\right) = \frac{-i_{avg}}{i_o}$$

$$\therefore \left.\frac{\partial i}{\partial \eta_s}\right|_{i_{avg}} = \frac{\alpha_c F \; i_o}{RT} \cdot \left(\frac{-i_{avg}}{i_o}\right) = -\frac{\alpha_c F}{RT} i_{avg}$$

$$R_{ct} = \left(\left.\frac{\partial i}{\partial \eta_s}\right|_{i=i_{avg}}\right)^{-1} = \boxed{\frac{RT \, i_{avg}}{-\alpha_c F} = R_{ct}}$$ ← Rct for Cathodic Tafel

For B-V kinetics:

$$i = i_o \left[\exp\left(\frac{\alpha_a F \eta_s}{RT}\right) - \exp\left(\frac{-\alpha_c F \eta_s}{RT}\right)\right] \Rightarrow \frac{i_{avg}}{i_o} = \exp\left(\frac{\alpha_a F \eta}{RT}\right) - \exp\left(\frac{\alpha_c F \eta}{RT}\right)$$

$$\frac{\partial i}{\partial \eta_s} = i_o \left(\frac{\alpha_a F}{RT}\right) \exp\left(\frac{\alpha_a F \eta_s}{RT}\right) - i_o \left(\frac{-\alpha_c F}{RT}\right) \exp\left(\frac{-\alpha_c F \eta_s}{RT}\right)$$

$$= i_o \left(\frac{F}{RT}\right) \left[\alpha_a \exp\left(\frac{\alpha_a F \eta_s}{RT}\right) + \alpha_c \exp\left(\frac{-\alpha_c F \eta_s}{RT}\right)\right]$$

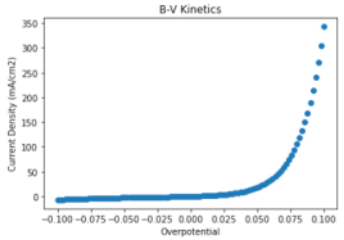
$$= \frac{F i_o}{RT} \left[\alpha_a \left(\frac{i_{avg}}{i_o} + \exp\left(\frac{\alpha_c F \eta_s}{RT}\right)\right) + \alpha_c \exp\left(\frac{-\alpha_c F \eta_s}{RT}\right)\right]$$

$$= \frac{F i_o}{RT} \left[\alpha_a \frac{i_{avg}}{i_o} + (\alpha_a + \alpha_c) \exp\left(\frac{-\alpha_c F \eta_s}{RT}\right)\right]$$

$$\boxed{R_{ct} = \left(\left.\frac{\partial i}{\partial \eta}\right|_{i=i_{avg}}\right)^{-1} = \frac{RT}{F i_o \left[\alpha_a \exp\left(\frac{\alpha_a F \eta}{RT}\right) + \alpha_c \exp\left(\frac{-\alpha_c F \eta}{RT}\right)\right]}}$$ ↙ BV kinetics R_{ct}

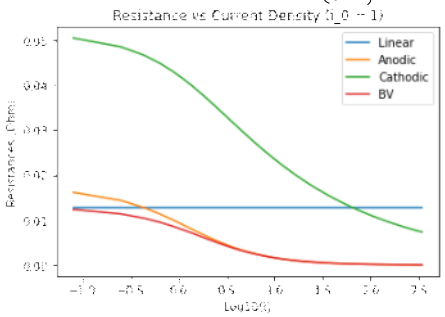
(b) Graphs shown

Plotting BV (i vs. η)

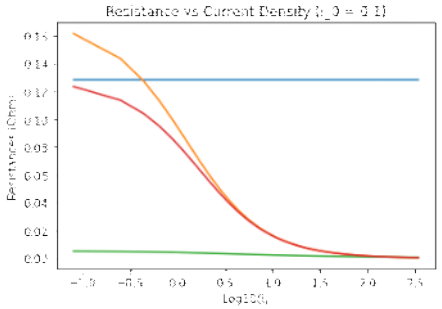


(c) Resistance vs $\log(i_0)$

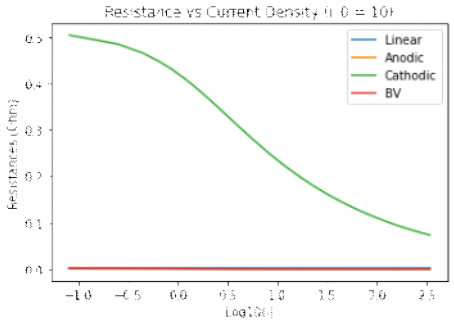
($i_0 = 1$)



($i_0 = 0.1$)



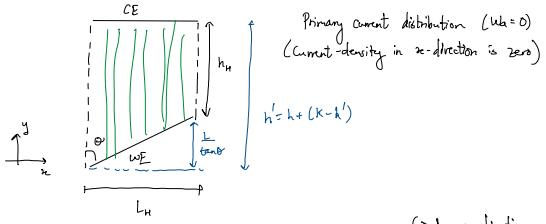
$i_0 = 10$



From the above graphs, it seems that Anodic Tafel approximation matches B-V the closest; Linear may be accurate only in small current densities but is independent of i
Cathodic diverges at first but then approaches B-V / Anodic over higher i_0 values
At higher i_0, the differences reduce (apart from cathodic)

Question 3 (W-6.3)

Hull cell



Primary current distribution ($\kappa_a = 0$)
(Current-density in x-direction is zero)

Current density only in y-direction : $\nabla^2\phi = 0$ (Only y-direction needed)

$i = -\kappa \nabla\phi$

only y-dependence

$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0$

$\frac{\partial^2\phi}{\partial y^2} = 0$

$\frac{\partial\phi}{\partial y} = c_1$

$\phi = c_1 y + c_2$

BC's:
(i) @ $y = h'$, $\phi = 0$
@ $y = K - k'$, $\phi = V$

$\Rightarrow$ $V = c_1(K - k') + c_2$
(-) $0 = c_1 h' + c_2$
$V = c_1[K - k' - h']$

$c_1 = \frac{V}{[K - k' - h']} = \frac{V}{[K - k - K + k']} = \frac{V}{-h}$

$i = -\kappa \nabla\phi = -\kappa \frac{\partial\phi}{\partial y}$

$i = +\kappa \frac{V}{h}$

$i = \frac{\kappa V}{h}$

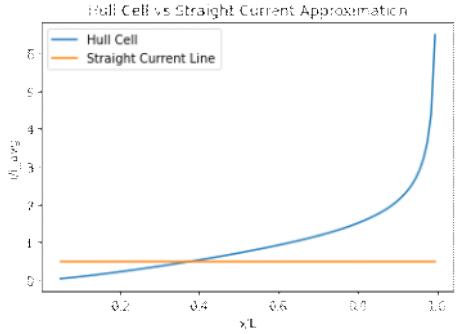
$i_{avg} = \int_0^L i \cdot dx = \frac{\kappa V}{h} \cdot L$

$\therefore$ $\boxed{\frac{i}{i_{avg}} = \frac{\kappa V}{h} \cdot \frac{h}{\kappa V L} = \frac{1}{L}}$

; $i_{avg} = \frac{\kappa V}{(h + \frac{L}{\tan\theta})}$ } Total "y" length

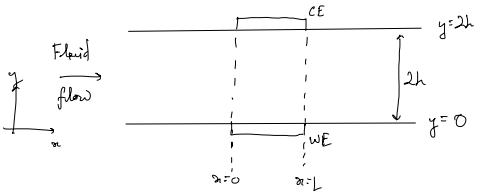
$\frac{i}{i_{avg}} = \frac{1}{h} \cdot \left(h + \frac{L}{\tan\theta}\right) = 1 + \frac{L}{h}\frac{1}{\tan\theta} = \frac{i}{i_{avg}}$

Plotting $\frac{i}{i_{avg}}$ for $\frac{L}{L} = 0.7385$ & $\theta = 0.22\pi$ we get the following graphical representation:



Question 4

Microfluidic eDean cell; 2 line electrodes



Equation 45: $\frac{3U}{h} y \frac{\partial c_{lim}}{\partial x} = D_{lim} \frac{\partial^2 c_{lim}}{\partial y^2}$

Similarity transformation: $\xi = y \left(\frac{U}{2hD_{lim} x} \right)^{1/3}$

$$\xi^3 = y \left(\frac{U}{2hD_{lim} x} \right) \Rightarrow \frac{U_y}{h} = \xi^3 \cdot 3 D_{lim} x$$

As shown in class,

$$\frac{\partial c}{\partial x} = \frac{\partial c}{\partial \xi} \cdot \frac{\partial \xi}{\partial x} \quad ; \quad \frac{\partial \xi}{\partial x} = y \left(\frac{U}{2hD} \right)^{1/3} \left(-\frac{1}{3} \right) x^{-4/3} = \frac{-\xi}{3x}$$

$$\rightarrow \frac{\partial c}{\partial x} = \frac{\partial c}{\partial \xi} \cdot \left(\frac{-\xi}{3x} \right)$$

Similarly,

$$\frac{\partial c}{\partial y} = \frac{\partial c}{\partial \xi} \cdot \frac{\partial \xi}{\partial y} \quad ; \quad \frac{\partial \xi}{\partial y} = \left(\frac{U}{3hDx} \right)^{1/3} = \frac{\xi}{y}$$

$$\frac{\partial c}{\partial y} = \left(\frac{\xi}{y} \right) \frac{\partial c}{\partial \xi}$$

We need: $\frac{\partial^2 c}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial c}{\partial y} \right) = \frac{\partial}{\partial y} \left(\frac{\xi}{y} \frac{\partial c}{\partial \xi} \right) = \frac{\xi}{y} \frac{\partial}{\partial y} \left(\frac{\partial c}{\partial \xi} \right) + \frac{\partial c}{\partial \xi} \frac{\partial}{\partial y} \left(\left[\frac{U}{2hDx} \right]^{1/3} \right)$ (the last term $\rightarrow 0$)

$$\frac{\partial^2 c}{\partial y^2} = \frac{\xi}{y} \left(\frac{\partial}{\partial \xi} \frac{\partial \xi}{\partial y} \frac{\partial c}{\partial \xi} \right)$$

$$= \frac{\partial^2 c}{\partial \xi^2} \cdot \left(\frac{\partial \xi}{\partial y} \right)^2$$

$$\rightarrow \frac{\partial^2 c}{\partial y^2} = \frac{\partial^2 c}{\partial \xi^2} \cdot \left(\frac{U}{3hDx} \right)^{2/3}$$

Plugging into original PDE:

$$\frac{3U}{h} y \frac{\partial c}{\partial \xi} \left(\frac{-\xi}{3x} \right) = D \frac{\partial^2 c}{\partial \xi^2} \left(\frac{U}{3hDx} \right)^{2/3} \Big) \div \left(\frac{U}{3hDx} \right)^{2/3}$$

$$3y \left(\frac{U}{3hDx} \right)^{1/3} \frac{\partial c}{\partial \xi} (-\xi) = \frac{\partial^2 c}{\partial \xi^2}$$

$$\therefore \quad \frac{\partial^2 c}{\partial \xi^2} + 3\xi^3 \frac{\partial c}{\partial \xi} = 0 \xrightarrow{\text{Since only change w.r.t 1 var}} \boxed{\frac{d^2c}{d\xi^2} + 3\xi^3 \frac{dc}{d\xi} = 0}$$

BCs: @ $x=0$, $c_{lim} = c_\infty \Rightarrow \xi \to \infty$ $\quad c_{lim} = c_\infty$

@ $y=\infty$, $c_{lim} = c_\infty \Rightarrow \xi \to \infty$ $\quad c_{lim} = c_\infty$

@ $y=0$, $c_{lim} = 0 \Rightarrow @\xi = 0$ $\quad c_{lim} = 0$

Using Forbenius solution: $\quad \frac{c_{lim}}{c_\infty} = \frac{1}{\Gamma(4/3)} \int_0^\xi e^{-x^3} \cdot dx$

$$N_y = -D\frac{\partial c}{\partial y}\bigg|_{y=0} = -D\frac{\partial c}{\partial \xi}\bigg|_{\xi=0} \frac{\partial \xi}{\partial y}\bigg|_{y=0}$$

$$N_y = \frac{-Dc_\infty}{\Gamma(4/3)} \underbrace{\frac{\partial}{\partial \xi}\left(\int_0^\xi e^{-x^3} dx \right)\bigg|_{\xi=0}}_{\text{F.T.C}} \left(\frac{U}{3hDx}\right)^{1/3}$$

$$N_y = \frac{-Dc_\infty}{\Gamma(4/3)} e^{-\xi^3}\bigg|_{\xi=0} = \frac{-c_\infty D}{\Gamma(4/3)} \left(\frac{U}{3hDx}\right)^{1/3}$$

Find Current $\longrightarrow$ $\quad \frac{si}{nF} = -N_y$

$$\frac{si}{nF} = \frac{Dc_\infty}{\Gamma(4/3)} \left(\frac{U}{3hDx}\right)^{1/3}$$

$$i_{avg} = \left(\frac{nF}{s} \frac{Dc_\infty}{\Gamma(4/3)} \left(\frac{U}{3hD}\right)^{1/3} \int_0^L x^{-1/3} \cdot dx \right) \frac{1}{L}$$

$$i_{avg} = \frac{nF}{s} \left(\frac{3}{2} \frac{D^{2/3}}{\Gamma(4/3)} \right) c_\infty \left(\frac{U}{3h}\right)^{1/3} L^{-1/3}$$

$-\frac{1}{3} + 1^{1/3} = \frac{2}{3}$

$\frac{L^{2/3}}{L} = L^{-1/3}$

$$\frac{i}{i_{avg}} = \frac{\frac{nF}{s} \frac{D^{1/3} c_\infty}{\Gamma(4/3)} \left(\frac{U}{3hD}\right)^{1/3} (x)^{-1/3}}{\frac{nF}{s} \frac{D^{1/3} c_\infty}{\Gamma(4/3)} \left(\frac{3}{2}\right) \left(\frac{U}{3hD}\right)^{1/3} \cdot L^{-1/3}}$$

$$\therefore \quad \boxed{\frac{i}{i_{avg}} = \frac{2}{3}\left(\frac{L}{x}\right)^{1/3}}$$

Question 5

$$\delta = \frac{-Dc_\infty}{N} \quad ; \quad \frac{\delta}{\delta_{avg}} = \frac{\frac{-Dc_\infty}{N}}{\frac{-Dc_\infty}{N_{avg}}} = \frac{N_{avg}}{N}$$

From previous part, $N_y = \frac{-c_\infty D}{\Gamma(4/3)} \left(\frac{U}{3hDx}\right)^{1/3}$

$$\therefore \quad \delta = \frac{-Dc_\infty}{-c_\infty D} \Gamma(4/3) \left(\frac{U}{3hD}\right)^{-1/3} \cdot x^{1/3}$$

$$\delta_{avg}(x) = \left(\int_0^L \Gamma(4/3) \left(\frac{U}{3hD}\right)^{-1/3} x^{1/3} \right) \frac{1}{L}$$

$$= \Gamma(4/3) \left(\frac{U}{3hD}\right)^{-1/3} \cdot \left(\frac{3}{4}\right) x^{4/3}\Big|_0^L \cdot \frac{1}{L} \qquad \frac{4}{3} - 1^{1/3} = \frac{1}{3}$$

$$= \Gamma(4/3) \left(\frac{U}{3hD}\right)^{-1/3} \left(\frac{3}{4}\right) L^{1/3}$$

$$N_y = -\frac{D}{\Gamma(4/3)} \left(\frac{U}{3hD}\right)^{1/3} \frac{c_\infty}{(x^{1/3})} = -\underbrace{\left(\frac{U^{1/3}}{(3h)^{1/3}} \cdot \frac{1}{\Gamma(4/3)}\right)}_{A} \frac{D^{2/3} c_\infty}{x^{1/3}}$$

$$N_{y,avg} = \left(-\frac{D}{\Gamma(4/3)} \left(\frac{U}{3hD}\right)^{1/3} c_\infty \int_0^L x^{-1/3} \cdot dx \right) \frac{1}{L}$$

$$N_{y,avg} = -\frac{D^{2/3}}{\Gamma(4/3)} \left(\frac{U}{3h}\right)^{1/3} c_\infty L^{-1/3} \left(\frac{3}{2}\right)$$

$$= -\underbrace{\left(\frac{U^{1/3}}{(3h)^{1/3}} \frac{1}{\Gamma(4/3)}\right)}_{A} \frac{D^{2/3} c_\infty}{L^{1/3}} \left(\frac{3}{2}\right) = \frac{3}{2} \frac{A D^{2/3} c_\infty}{\delta_{avg}}$$

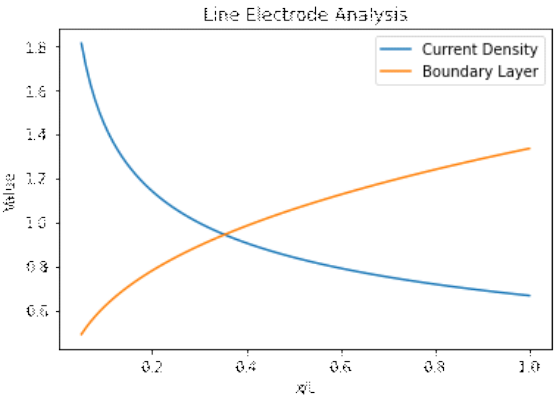
$$\therefore \quad \frac{\delta(x)}{\delta_{avg}} = \frac{N_{y,avg}}{N_y} = \frac{-A D^{2/3} c_\infty \left(\frac{3}{2}\right) / L^{1/3}}{-A D^{2/3} c_\infty} = \boxed{\frac{3}{2}\left(\frac{x}{L}\right)^{1/3} = \frac{\delta(x)}{\delta_{avg}}}$$

$$= \Gamma(4/3)\left(\frac{U}{3hD}\right)^{-1/3}\left(\frac{3}{4}\right)^{1/3} L^{1/3}$$

$$\therefore \frac{S}{S_{avg}} = \frac{\Gamma(4/3)\left(\frac{U}{3hD}\right)^{-1/3} x^{1/3}}{\Gamma(4/3)\left(\frac{U}{3hD}\right)^{-1/3}\left(\frac{3}{4}\right)^{1/3} L^{1/3}} = \boxed{\frac{4}{3}\left(\frac{x}{L}\right)^{1/3} = \frac{S}{S_{avg}}}$$

$$\therefore \frac{S(x)}{S_{avg}} = \frac{N_{y,avg}}{N_y} = \frac{-A D^{2/3} C_\infty \left(\frac{3}{2}\right)}{\frac{L^{1/3}}{-A D^{2/3} C_\infty}} = \boxed{\frac{3}{2}\left(\frac{x}{L}\right)^{1/3} = \frac{S(x)}{S_{avg}}}$$

Plotting $\frac{i}{i_{avg}}$ & $\frac{S}{S_{avg}}$



Question 6

Levich plot @ specific overpotential → Find diffusion coefficient

$$m = 0.62\left(\frac{nFC_\infty}{S_{lim}}\right)\nu^{-1/6} D^{2/3}$$

$$\left[\frac{m}{0.62}\nu^{1/6}\left(\frac{S_{lim}}{nFC_\infty}\right)\right]^{\frac{3}{2}} = D$$

$$\frac{mA}{cm^2} = \frac{10^{-3}}{(10^{-2})^2}\frac{A}{m^2}$$

C_∞ → conc. of H_2 in the bulk solution

$$\frac{0.00016\ g\ H_2}{100\ g\ H_2O} \approx \frac{0.00016\ g\ H_2}{100\ mL\ H_2O} = \frac{0.00016 \times \frac{1}{2.02}}{100 \times 10^{-6}} = 0.792\ \frac{mol}{m^3}$$

ν → kinematic viscosity (can be approx. close to water since 0.1M K_2SO_4 used)

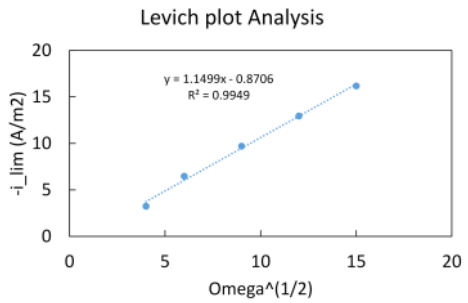
From Engineering toolbox: $\nu = 8.92 \times 10^{-7}\ m^2/s$

$n=2$, $F = 96485$, $S_{lim} = 2$

From the levich plot: slope = 1.1499

$$\therefore \boxed{D = 3.67 \times 10^{-5}\ cm^2/s}$$

$D_{lit} = 4.58 \times 10^{-6}\ cm^2/s$ } within a factor of 0.8 !



Question 7

Koutecky-Levich Analysis ; Magnitude of charge transfer resistance vs. MT resistance in system.

From the plot of $\frac{1}{i_{lim}}$ vs. $\frac{1}{\Omega^{1/2}}$ we get: $m = 0.7301$, $c = 0.0325$

$$m = \frac{1}{(0.62) D^{2/3} \nu^{-1/6} C_\infty \cdot F \cdot n}$$

$$\therefore \frac{1}{i_{lim}} = \frac{1}{i_K} + \frac{1}{i_D} \quad ; \quad \frac{1}{i_K} = 0.0325 \Rightarrow i_K = 30.77\ \frac{A}{m^2}$$

$$D^{2/3} = \frac{1}{(0.62) m \cdot \nu^{-1/6} C_\infty \cdot F \cdot n}$$

$$D = \left(\frac{1}{(0.62) m \nu^{-1/6} C_\infty \cdot F \cdot n}\right)^{3/2}$$

For 100 rpm: $\frac{1}{i_{lim}} = 0.0325$

i_{lim} i_K i_D i_K $\frac{A}{m^2}$

$D = \left(\frac{1}{(0.12)\, m\, \nu^{-1/6}\, c_0 \cdot F \cdot n}\right)^{3/2}$

For 100 rpm: $\frac{1}{i_K} = 0.0325$

$\frac{1}{i_D} = 0.2175$

2500 rpm: $\frac{1}{i_K} = 0.0325$

$\frac{1}{i_D} = 0.03417$

Similar to summing up resistors:

→ As rpm ↑, mass transfer resistance decreases (which charge transfer resistance stays the same)

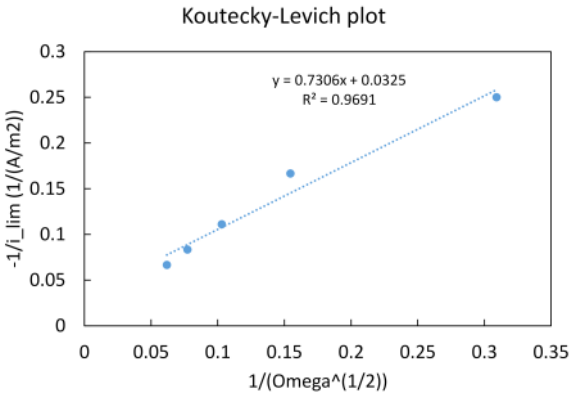
→ Charge transfer resistance always smaller than mass transfer resistance (i.e. always in diffusion limitations for the range of rpms in the demo)

$D = 1.68 \times 10^{-5}\ cm^2/s$

$D_{Lit} = 4.58 a 10^{-5}\ cm^2/s$

These are off by a factor of 2.73

↳ possibly due to contributions for ORR occuring simultaneously



Question 8

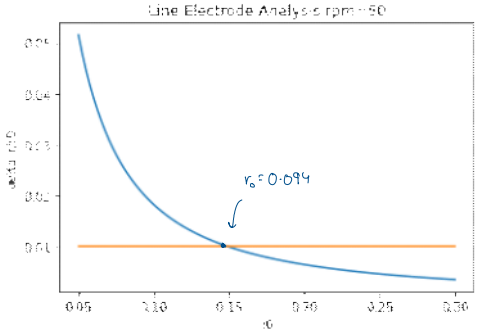
(a) $\Delta r = 0\left(\sqrt{\frac{D\nu^{1/2}}{\Omega^{3/2} r_0}}\right)$; $D = 10^{-5}\ cm^2 s^{-1}$, $\nu = 10^{-2}\ cm^2 s^{-1}$

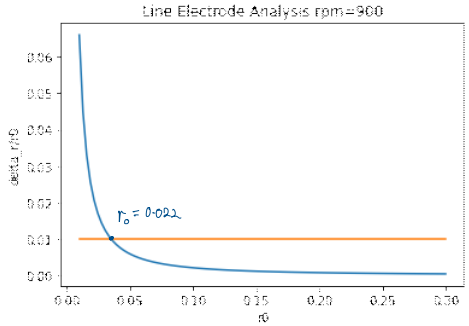
rpm → rad/s

$1\ rpm = \frac{1\ rotation}{1\ min} = \frac{2\pi}{60}\ \frac{rad}{s}$

Plotted graphs allow to calculate the required r_0 value

rpm ↑ → r_0 ↓ (Smaller r_0 needed to go below 0.01 ratio)





(b) Solving for Δr order given r_0, Ω, & ν gives:

(b) solving for Δr order given r_0, Ω, & v gives:

$$\Delta r^{100\ rpm} = 7.67 \times 10^{-4}$$

$$\Delta r^{3600\ rpm} = 5.21 \times 10^{-5}$$

↓ Higher rpm reduces the edge effect region